LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Practical demonstration of steady streamlined flow pattern over object in a smoke tunnel.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Practical demonstration of steady streamlined flow pattern over object in a smoke tunnel.
5. Specific Objectives	6.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of steady streamlined flow pattern over object in smoke tunnel; Safety: Ensure smoke density is controlled; Supervision: Instructor monitors trainee observation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three primary properties of the atmosphere affecting flow?
(b) Define the term 'boundary layer' in fluid dynamics.
(c) List two factors that influence flow separation over a body.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Properties of Atmosphere • Flow over Bodies • Boundary Layer Formation • Streamlined vs Turbulent Flow • Smoke Tunnel Mechanics	L & Ds	PPT, IB	15 min
(b)	• Setup of Smoke Tunnel • Introduction of Test Object • Generation of Smoke Stream • Observation of Flow Patterns • Identification of Separation Points	L & Ds	PPT, PDF	20 min
(c)	• Analysis of Streamlines • Effect of Object Shape • Reynolds Number Implications • Recording Observations • Comparison with Theory	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the boundary layer affects the transition from streamlined to turbulent flow.
(b) Describe the visual characteristics of a steady streamlined flow pattern in the smoke tunnel.

12. Home Assignment and Follow Up	Prepare a brief report comparing the observed flow patterns in this session with the theoretical predictions covered in LP 02 regarding fluid dynamics.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Video demonstrating fluid flow characteristics. Video on theory of flight Lift (b) Drag (c) control.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min